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1  /* Create a comprehensive hospital performance dashboard using CTEs. Calculate:
2     1) Service-level metrics (total admissions, refusals, avg satisfaction),
3     2) Staff metrics per service (total staff, avg weeks present),
4     3) Patient demographics per service (avg age, count).
5     Then combine all three CTEs to create a final report showing service name,
6     all calculated metrics, and an overall performance score (weighted average of admission rate and satisfaction).
7     Order by performance score descending. */
8  • WITH
9  all_services AS (
10     SELECT DISTINCT service FROM services_weekly
11     UNION
12     SELECT DISTINCT service FROM patients
13     UNION
14     SELECT DISTINCT service FROM staff
15 ),
16 services_weekly_agg AS (
17     SELECT
18         sw.service,
19         SUM(sw.patients_admitted)    AS sw_total_admitted,
20         SUM(sw.patients_refused)     AS sw_total_refused,
21         ROUND(AVG(sw.patient_satisfaction), 2) AS sw_avg_satisfaction,
22         ROUND(AVG(sw.staff_morale), 2)      AS sw_avg_staff_morale,
23         ROUND(AVG(sw.available_beds), 2)    AS sw_avg_available_beds,
24         ROUND(AVG(sw.patients_request), 2)  AS sw_avg_patients_request,
25         ROUND(AVG(sw.event IS NOT NULL), 2) AS sw_event_rate -- fraction of weeks with an event (0 or 1 per row)
26     FROM services_weekly sw
27     GROUP BY sw.service
28 ),

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29  patient_service AS (  
30      SELECT p.service, COUNT(*) AS total_admissions_from_patients, ROUND(AVG(p.satisfaction), 2) AS avg_satisfaction_from_patients  
31      FROM patients p  
32      GROUP BY p.service  
33  ),  
34  staff_presence_per_staff AS (  
35      SELECT ss.staff_id, ss.service, SUM(CASE WHEN ss.present = 1 THEN 1 ELSE 0 END) AS weeks_present  
36      FROM staff_schedule ss  
37      GROUP BY ss.staff_id, ss.service  
38  ),  
39  staff_metrics AS (  
40      SELECT  
41          s.service, COUNT(DISTINCT s.staff_id) AS total_staff,  
42          ROUND(AVG(COALESCE(sp.weeks_present, 0)), 2) AS avg_weeks_present  
43      FROM staff s  
44      LEFT JOIN staff_presence_per_staff sp  
45          ON s.staff_id = sp.staff_id  
46          AND s.service = sp.service  
47      GROUP BY s.service  
48  ),  
49  patient_demo AS (  
50      SELECT p.service, ROUND(AVG(p.age), 2) AS avg_age, COUNT(*) AS patient_count  
51      FROM patients p  
52      GROUP BY p.service  
53  )
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54 SELECT
55     a.service AS service_name, COALESCE(sw.sw_total_admitted, 0) AS sw_total_admitted,
56     COALESCE(sw.sw_total_refused, 0) AS sw_total_refused,
57     sw.sw_avg_satisfaction AS sw_avg_satisfaction,
58     sw.sw_avg_staff_morale AS sw_avg_staff_morale,
59     sw.sw_avg_available_beds AS sw_avg_available_beds,
60     sw.sw_avg_patients_request AS sw_avg_patients_request,
61     sw.sw_event_rate AS sw_event_rate,
62     COALESCE(ps.total_admissions_from_patients, 0) AS total_admissions_from_patients,
63     ps.avg_satisfaction_from_patients AS avg_satisfaction_from_patients,
64     COALESCE(sm.total_staff, 0) AS total_staff,
65     COALESCE(sm.avg_weeks_present, 0) AS avg_weeks_present,
66     COALESCE(pd.avg_age, 0) AS avg_age,
67     COALESCE(pd.patient_count, 0) AS patient_count,
68     ROUND(CASE WHEN NULLIF( COALESCE(sw.sw_total_admitted,0) + COALESCE(sw.sw_total_refused,0), 0) IS NULL
69         THEN NULL ELSE COALESCE(sw.sw_total_admitted,0) / NULLIF( COALESCE(sw.sw_total_admitted,0) + COALESCE(sw.sw_total_refused,0), 0)END, 4) AS admission_rate,
70     -- consistency metric: ratio of patient-table admissions to services_weekly admitted totals
71     ROUND(CASE WHEN NULLIF(COALESCE(sw.sw_total_admitted,0), 0) IS NULL THEN NULL
72         ELSE COALESCE(ps.total_admissions_from_patients,0) / NULLIF(COALESCE(sw.sw_total_admitted,0),0)
73         END, 4) AS admission_coverage_ratio,
74     -- performance score (weights: 60% admission_rate, 40% average satisfaction)
75     ROUND(0.6 * COALESCE(CASE
76         WHEN NULLIF( COALESCE(sw.sw_total_admitted,0) + COALESCE(sw.sw_total_refused,0), 0) IS NULL
77         THEN 0 ELSE COALESCE(sw.sw_total_admitted,0) / NULLIF( COALESCE(sw.sw_total_admitted,0) + COALESCE(sw.sw_total_refused,0), 0) END, 0) + 0.4 *
78         (COALESCE(sw.sw_avg_satisfaction, COALESCE(ps.avg_satisfaction_from_patients,0)) / 100.0 ), 4) AS performance_score
79 FROM all_services a
80 LEFT JOIN services_weekly_agg sw ON a.service = sw.service
81 LEFT JOIN patient_service ps ON a.service = ps.service
82 LEFT JOIN staff_metrics sm ON a.service = sm.service
83 LEFT JOIN patient_demo pd ON a.service = pd.service
84 ORDER BY performance_score DESC, a.service;

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